CHAPTER 1

FUNDAMENTALS OF COMPUTER

CHAPTER ESSENCE AT A GLANCE:
Modern electronic computers transform raw unorganized facts into contextual decision-making information through synchronized execution across the central processing unit, stratified memory, and peripheral interfaces.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Data Processing Cycle and Information Dynamics:** Data consists of unorganized atomic facts, figures, and symbols lacking contextual relevance on their own, whereas information represents structured, processed data organized into meaningful context to support human decision making.

• **Von Neumann Architecture and Functional Units:** The stored-program model integrates an input unit for data ingestion, a central processing unit containing an arithmetic logic unit and control unit for execution and timing, primary memory for simultaneous storage of instructions and operands, and an output unit for information delivery.

• **Hierarchical Storage Stratification and Latency:** Computer storage balances access latency against economic cost and capacity, cascading from high-speed CPU registers and SRAM cache memory down to volatile DRAM main memory and high-capacity non-volatile solid state and magnetic secondary storage media.

• **Binary Digitization and Universal Character Encoding:** All text, numbers, and multimedia instructions are represented internally as binary digits, organized into positional number systems and encoded globally using ASCII and multi-byte Unicode UTF-8 standards.

ESSENTIAL ARCHITECTURE & CORE SYNTAX CARDS

$$\text{System Throughput} = \frac{\text{Input Stream} \xrightarrow{\text{Data}} \text{CPU (ALU + CU + Registers)} \xrightarrow{\text{Info}} \text{Output}}{\text{Primary Memory (RAM + ROM)} \longleftrightarrow \text{Secondary Storage (SSD + HDD)}}$$

Von Neumann Functional Architecture and Data Flow

$$N_b = \sum_{i=0}^{n-1} d_i \cdot b^{\,i} + \sum_{j=1}^{m} d_{-j} \cdot b^{\,-j} \quad (\text{Radix } b \in \{2, 8, 10, 16\})$$

Positional Number System Expansion Theorem

$$1\ \text{Byte} = 8\ \text{Bits}, \quad 1\ \text{KB} = 2^{10}\ \text{Bytes}, \quad 1\ \text{MB} = 2^{20}\ \text{Bytes}, \quad 1\ \text{GB} = 2^{30}\ \text{Bytes}$$

Binary Storage Capacity Scaling Units

$$\text{Latency Hierarchy} = \frac{\text{Registers } (\sim 1\ \text{ns}) > \text{SRAM Cache } (\sim 2 \text{ to } 10\ \text{ns}) > \text{DRAM Main } (\sim 50\ \text{ns})}{\text{Solid State SSD } (\sim 100\ \mu\text{s}) > \text{Magnetic HDD } (\sim 10\ \text{ms})}$$

Access Latency across Memory and Storage Tiers

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Never confuse raw data with processed information; data represents unorganized atomic facts while information provides contextual meaning for decision making.

• **Alert:** Remember that RAM is volatile working memory wiped clean upon power failure, whereas ROM is non-volatile permanent memory preserving boot firmware BIOS.

• **Alert:** When converting fractional decimal numbers to binary, use successive multiplication by 2 and collect the carry integers from top to bottom.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Distinguish between data and information with a suitable real-world example. [Score: 2 Marks]

Answer & Key Valuation Steps:

• Data: Raw, unorganized, and unprocessed facts, figures, or observations that lack contextual meaning on their own (e.g. marks obtained by students: 45, 88, 92). **[1 Mark]**

• Information: Processed, organized, and structured data presented in a meaningful context that aids in understanding and decision making (e.g. Average class mark: 75, Grade sheet). **[1 Mark]**

2. Explain the functional units of a computer system based on the Von Neumann stored-program model. [Score: 2 Marks]

Answer & Key Valuation Steps:

• Input and Output Units: The Input Unit accepts external data and converts it into machine-readable binary signals; the Output Unit converts processed binary results into human-comprehensible visual or printed formats. **[1 Mark]**

• Central Processing Unit (CPU) and Memory: CPU contains the Arithmetic Logic Unit (ALU) for computations, the Control Unit (CU) for timing and coordination, and Registers for high-speed storage, working synchronously with Primary Memory. **[1 Mark]**

3. Convert the decimal number 45.625 into its equivalent binary number system representation. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Integer Part Conversion: Divide 45 successively by 2. 45 divided by 2 is 22 remainder 1; 22 divided by 2 is 11 remainder 0; 11 divided by 2 is 5 remainder 1; 5 divided by 2 is 2 remainder 1; 2 divided by 2 is 1 remainder 0; 1 divided by 2 is 0 remainder 1. Reading remainders bottom to top yields 101101. **[1.5 Marks]**

• Fractional Part Conversion: Multiply 0.625 successively by 2. 0.625 times 2 is 1.25 (carry 1); 0.25 times 2 is 0.50 (carry 0); 0.50 times 2 is 1.00 (carry 1). Reading carry values top to bottom yields 0.101. **[1 Mark]**

• Final Result: Combining integer and fractional parts gives (45.625) in decimal = (101101.101) in binary. **[0.5 Mark]**

4. Convert the binary number 1101101.101 into equivalent octal and hexadecimal numbers. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Conversion to Octal: Group bits in triads of 3 from the binary point. Integer part: (001)(101)(101) = 1 5 5. Fractional part: (101) = 5. Result = (155.5) base 8. **[1.5 Marks]**

• Conversion to Hexadecimal: Group bits in tetrads of 4 from the binary point. Integer part: (0110)(1101) = 6 D. Fractional part: (1010) = A. Result = (6D.A) base 16. **[1.5 Marks]**

5. Differentiate between Static RAM (SRAM) and Dynamic RAM (DRAM) with respect to speed, construction, and power. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Construction and Refreshing: SRAM is built using flip-flop transistor circuits and retains data continuously without refreshing; DRAM is constructed from capacitors and transistors where capacitors leak charge and demand periodic refreshing cycles. **[1 Mark]**

• Access Latency and Performance: SRAM provides ultra-low access times (2 to 10 ns) and serves as CPU cache; DRAM exhibits slower access latency (around 50 ns) and serves as main system memory. **[1 Mark]**

• Cost and Density: SRAM requires multiple transistors per bit making it expensive and lower density; DRAM requires one transistor and capacitor per cell, achieving massive gigabyte densities at affordable costs. **[1 Mark]**

6. Explain the specific functions of the Arithmetic Logic Unit (ALU), Control Unit (CU), and Registers inside the CPU. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Arithmetic Logic Unit (ALU): Performs basic mathematical computations (addition, subtraction, multiplication) and logical comparisons (equal to, greater than, boolean logic) on operands. **[1 Mark]**

• Control Unit (CU): Decodes software instructions, coordinates the instruction fetch-decode-execute cycle, and sends precise timing signals across the system to synchronize data flow. **[1 Mark]**

• Registers: Ultra-fast temporary storage locations located directly inside the CPU silicon holding active operands, instruction pointers, and status flags during execution. **[1 Mark]**

7. Describe ASCII and Unicode character coding standards. Explain why Unicode has emerged as the universal global standard. [Score: 4 Marks]

Answer & Key Valuation Steps:

• ASCII Coding Standard: American Standard Code for Information Interchange employs 7 bits (standard ASCII, 128 characters) or 8 bits (extended ASCII, 256 characters) to encode Latin alphabets, numbers, and punctuation. **[1 Mark]**

• Unicode Standard Architecture: Unicode defines a universal multi-byte encoding system (UTF-8, UTF-16, UTF-32) allocating over one million distinct code points to represent every living and historical language on Earth. **[1 Mark]**

• Limitations of Legacy ASCII: 8-bit ASCII is strictly limited to 256 symbols, making it incapable of encoding Indian languages (such as Malayalam, Hindi, Tamil) or Asian scripts simultaneously. **[1 Mark]**

• Global Computing Superiority: Unicode guarantees consistent multilingual data exchange, cross-platform software localization, and seamless web page rendering across all operating systems without character corruption. **[1 Mark]**